

Capstone Project: Full Alert-to-Response Cycle

- The objective of this capstone project was to simulate a Security Operations Center (SOC) workflow by performing cyberattack activities, generating security alerts, analyzing incidents, preserving evidence, and documenting the complete alert-to-response lifecycle using Wazuh SIEM.

- **Tool used :**

1. Kali linux
2. Wazuh
3. Metasploitable
4. Hydra
5. Nmap

Attack Simulation

- **Scanning a target**



```
(root@kali)-[/home/kali]
└─$ nmap -sV 10.186.27.32
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-15 11:02 +0530
Nmap scan report for 10.186.27.32
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:26:FA:D1 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.16 seconds

(root@kali)-[/home/kali]
```

- **Exploiting a vsftpd 2.3.4**

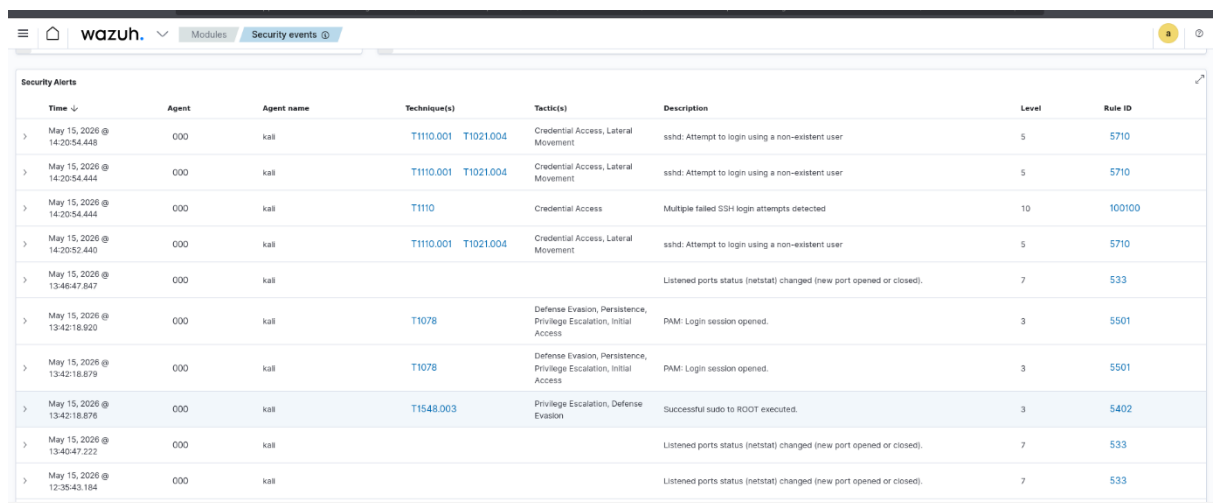
The Metasploit Framework was used to exploit the VSFTPD 2.3.4 backdoor vulnerability present in Metasploitable 2.

```
root@kali:/home/kali
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.186.27.32
RHOSTS => 10.186.27.32
msf exploit(unix/ftp/vsftpd_234_backdoor) > options
Module options (exploit/unix/ftp/vsftpd_234_backdoor):
  Name      Current Setting  Required  Description
  ----      -
  CHOST      no               no        The local client address
  CPORT      no               no        The local client port
  Proxies    no               no        A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     10.186.27.32    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      21               yes       The target port (TCP)

Exploit target:
  Id  Name
  --  -
  0    Automatic

View the full module info with the info, or info -d command.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.186.27.32
RHOSTS => 10.186.27.32
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 10.186.27.32:21 - Exploit failed [unreachable]: Rex::HostUnreachable The host (10.186.27.32:21) was unreachable.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 10.186.27.32:21 - Exploit failed [unreachable]: Rex::HostUnreachable The host (10.186.27.32:21) was unreachable.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 10.186.27.32:21 - Exploit failed [unreachable]: Rex::HostUnreachable The host (10.186.27.32:21) was unreachable.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

- **Alert Monitoring in Wazuh**



Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
May 15, 2026 @ 14:20:54.448	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
May 15, 2026 @ 14:20:54.444	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
May 15, 2026 @ 14:20:54.444	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100
May 15, 2026 @ 14:20:52.440	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
May 15, 2026 @ 13:46:47.847	000	kali			Listened ports status (netstat) changed (new port opened or closed).	7	533
May 15, 2026 @ 13:42:18.920	000	kali	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
May 15, 2026 @ 13:42:18.879	000	kali	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
May 15, 2026 @ 13:42:18.876	000	kali	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402
May 15, 2026 @ 13:40:47.222	000	kali			Listened ports status (netstat) changed (new port opened or closed).	7	533
May 15, 2026 @ 12:35:43.184	000	kali			Listened ports status (netstat) changed (new port opened or closed).	7	533

- **Triage Documentation**
- **TriagingTable**

Time stamp	Source ip	Alert description	MITRE technique
2026-05-14	10.186.27.32	VSFTPD exploit attempt	T1190

- **Capstone Incident Report**

- **Executive Summary**
 - A simulated exploitation attempt targeting the VSFTPD 2.3.4 backdoor vulnerability was conducted against the Metasploitable2 virtual machine using Metasploit. The activity was monitored through Wazuh SIEM,

enabling alert generation and security event analysis. The incident demonstrated the complete SOC workflow from detection to containment.

- **Timeline**

1. Vulnerability identified using Nmap service detection
2. Exploit executed using Metasploit framework
3. Wazuh detected suspicious FTP activity
4. Alert triage performed
5. Attacker IP isolated using firewall blocking rules
6. Investigation findings documented

- **Recommendations**

1. Disable vulnerable FTP services
2. Apply security patches regularly
3. Restrict unnecessary network access
4. Implement stronger SIEM correlation rules
5. Conduct continuous vulnerability scanning and monitoring

- **Stakeholder briefing**

A simulated cyberattack targeting a vulnerable FTP service was successfully detected and analyzed within the SOC environment. The security monitoring system identified suspicious activity associated with the exploit attempt, allowing the incident response process to begin immediately. The affected system was isolated by blocking the attacker IP address, preventing further unauthorized access. Investigation activities confirmed exploitation attempts against the vulnerable service, and relevant forensic evidence was preserved. Recommendations include patch management improvements, stronger monitoring rules, and reducing exposure of vulnerable services to minimize future attack risks.

- **Key Findings**

1. The Wazuh SIEM platform successfully collected and analyzed security events generated during the simulated attack activity in the SOC lab environment.
2. A brute-force SSH attack simulated using Hydra against the Metasploitable target generated detectable authentication failure events that were forwarded to Wazuh.
3. Wazuh correlated the authentication logs and generated alerts related to suspicious login attempts and repeated failed authentication activity.
4. Security events were mapped with relevant MITRE ATT&CK techniques to improve threat classification and investigation accuracy.
5. Alert triage activities helped distinguish legitimate security incidents from normal system behavior using severity levels and contextual analysis.

6. Incident response procedures including detection, analysis, containment planning, and documentation were successfully practiced.
7. Evidence preservation techniques such as log retention, alert export, timestamp verification, and event documentation were performed to maintain investigation integrity.
8. Dashboard monitoring in Wazuh provided centralized visibility into alerts, source IP addresses, rule levels, and attack behavior.
9. The project demonstrated the importance of continuous monitoring, alert prioritization, and incident documentation within SOC operations.

• Conclusion

The Capstone Project: Full Alert-to-Response Cycle provided practical exposure to real-world SOC operations and cybersecurity monitoring workflows. The project successfully demonstrated how security tools such as Wazuh SIEM can detect, analyze, and manage suspicious activities generated during simulated attacks.

By performing brute-force attack simulations using Hydra against the Metasploitable environment, the project showcased the complete lifecycle of security monitoring, including event generation, log collection, alert correlation, triage, investigation, and documentation.

The exercise also emphasized the importance of evidence preservation, incident response procedures, MITRE ATT&CK mapping, and alert prioritization in modern SOC environments. Through this project, practical understanding of security operations, threat detection, and defensive monitoring capabilities was significantly improved.

Overall, the capstone project strengthened both theoretical and hands-on cybersecurity skills relevant to SOC analysis, incident handling, and SIEM-based monitoring environments.

• References

1. Wazuh Documentation
2. [MITRE ATT&CK Framework](#)
3. [NIST SP 800-61 Incident Handling Guide](#)
4. [Hydra THC Documentation](#)
5. [Metasploitable2 Project](#)
6. [OpenSearch Documentation](#)
7. [SANS Incident Handler's Handbook](#)
8. [OWASP Security Logging Cheat Sheet](#)